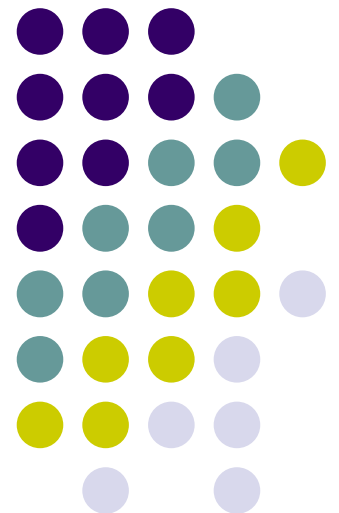


Multiple Access Channels

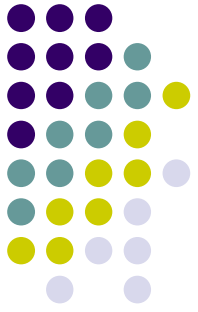
Some Queuing Theory

MAC: Aloha, ethernet

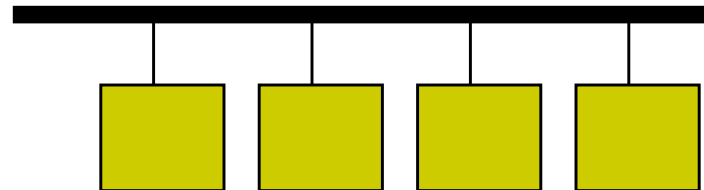
Exponential backoff & friends



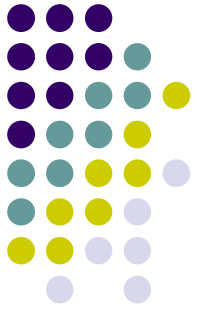
LANs: Local Area Networks



- Goal: extend benefits of simple connection as far as possible
- Means: Share medium between more than two endpoints
- Problem: how to access the medium?
- We'll look at a few approaches...



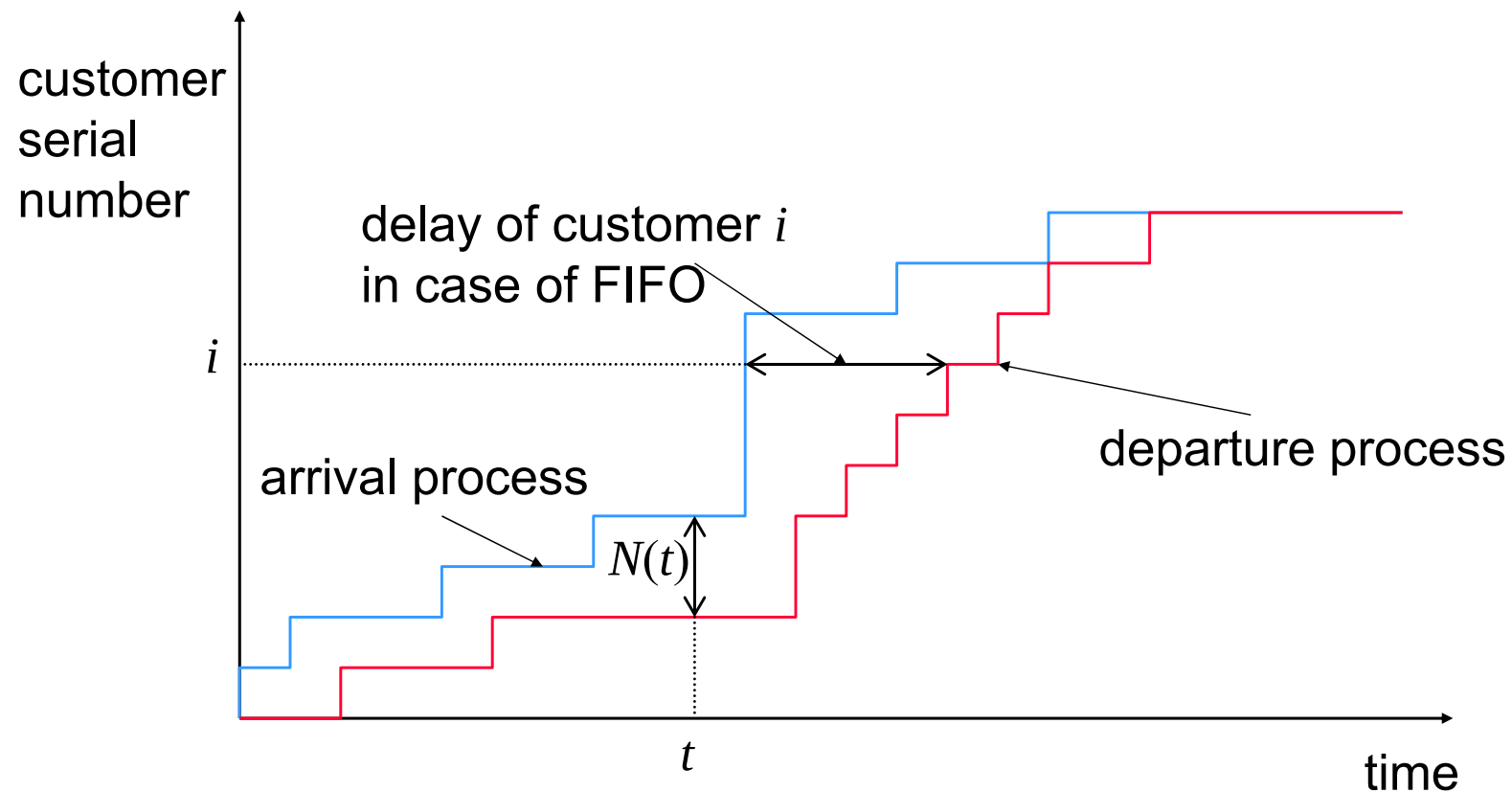
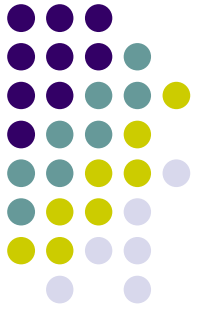
Alternative 1: Static Allocation



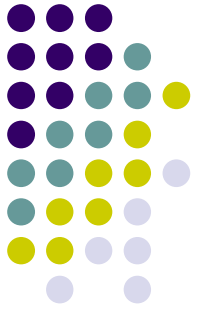
Approach:

- Divide the channel into n logical channels, let each node its own channel
- Use time, space or frequency division
 - Time (TDMA): user i get time slot j iff $j \equiv i \pmod{n}$
 - Space: each user has his own send line
 - Frequency (FDMA): divide frequency bandwidth
- How good is it?
 - We'll use **Queuing Theory** to study average waiting times.

Little's Theorem: Setting



Little's Theorem: Statement



Let

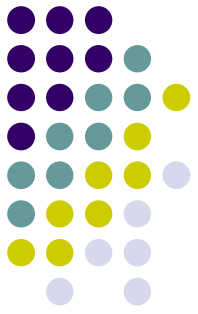
N : average # customers in system

λ : average arrival rate (in customers/time unit)

T : average delay in the system (in time units).

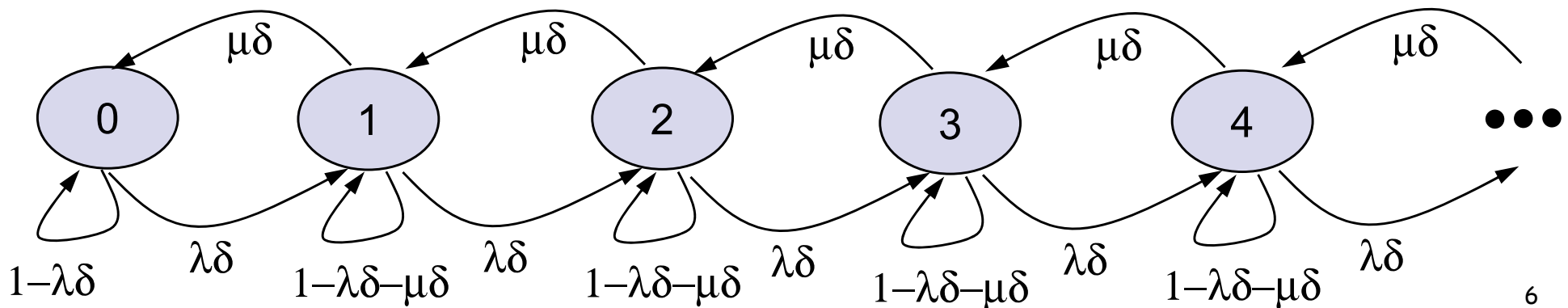
Then $N = \lambda \cdot T$; alternatively: $T = N/\lambda$.

Example: On average, people arrive 1 person/minute to the supermarket; there are 50 people in the supermarket. How much time, on average, does each one spend in the supermarket?

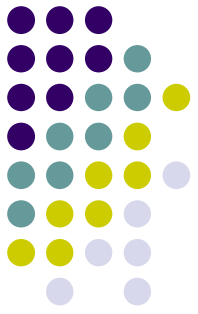


The M/M/1 Queuing Model

- Arrivals and departures are stochastic (prob.)
 - arrivals are Poisson process with rate λ (expcted interarrival time: $1/\lambda$)
 - service time is distributed exponentially with rate μ (average service time $1/\mu$).
- System state $\equiv$ number of customers
- Markov Chain: take a time window of length $\delta \rightarrow 0^+$

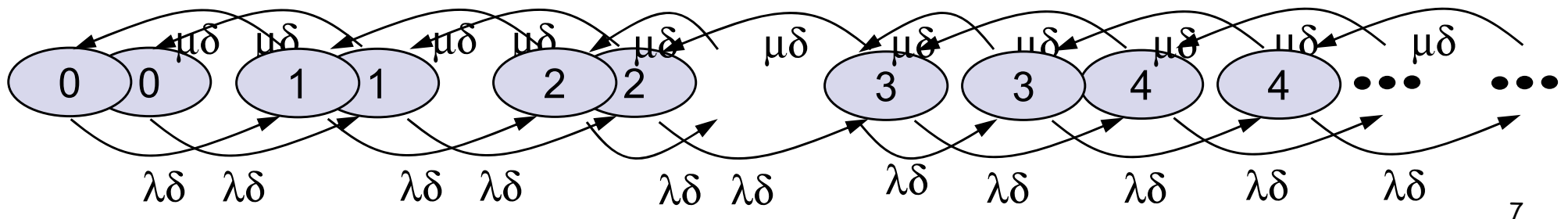


M/M/1 Steady State: state probabilities in the limit

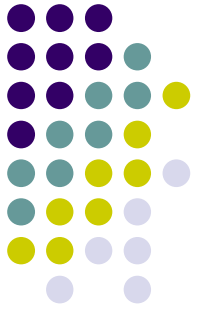


Def: $p_i = \lim_{time \rightarrow \infty} [\text{prob. of being in state } i]$

- In steady state: 0 net flow across each cut
- Therefore $p_i \mu \delta = p_{i-1} \lambda \delta \Rightarrow \dots p_i = p_0 \left(\frac{\lambda}{\mu} \right)^i$
- since also, $\sum_{i=0}^{\infty} p_i = 1$ we get $p_0 = 1 - \frac{\lambda}{\mu}$



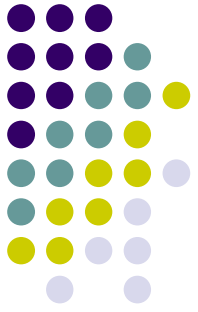
M/M/1 Average Queue Length



- Let $\rho = \lambda/\mu$ ("utilization factor")
- We saw that $p_i = p_0\rho^i$, and that $p_0 = 1-\rho$.
- So expected number of customers is

$$E[N] = \sum_{n=0}^{\infty} np_n$$

M/M/1 Average Waiting Time



Finally, we apply Little's theorem and find the expected delay:

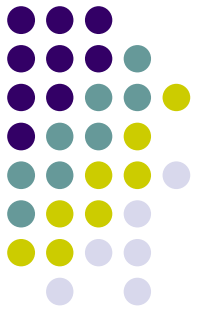
$$T = N/\lambda = \frac{\rho}{1 - \rho} \cdot \frac{1}{\lambda} = \frac{\lambda/\mu}{1 - \lambda/\mu} \cdot \frac{1}{\lambda} = \frac{1}{\mu - \lambda}$$

- Summary: In the M/M/1 model, where

- average arrival rate is λ pkts/sec, and
- average service rate is μ pkts/sec,

the average packet waiting time is $\frac{1}{\mu - \lambda}$ seconds

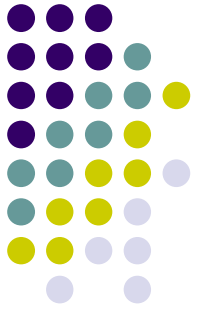
Back to multiple access channels: Static Allocation latency



Consider a channel with average arrival rate λ and average service rate μ

- If we divide the channel into n small channels, then for each small channel:
 - The arrival rate is λ/n
 - The service rate is μ/n
- So the average waiting time is

$$\frac{1}{\frac{\mu}{n} - \frac{\lambda}{n}} = \frac{n}{\mu - \lambda}$$



Implication of QT result

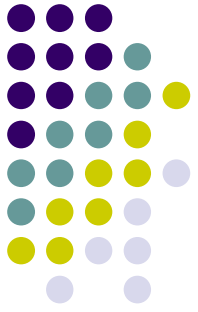
M/M/1 is a simplistic model, but if it's roughly correct, then...

- Even if splitting the channel is free (no overhead), the latency grows **linearly** with the splitting factor!

Conclusion: Don't statically split the channel, share it

- How?

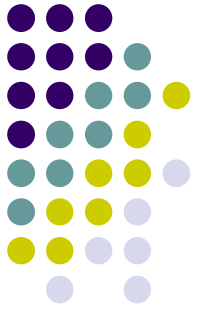
Medium Access Control (MAC) Protocols



- Goal: arbitrate access to a common shared channel among a population of users
- Desired properties:
 - Efficient
 - Low delay, high utilization
 - Fair
 - Fault tolerant (distributed!)

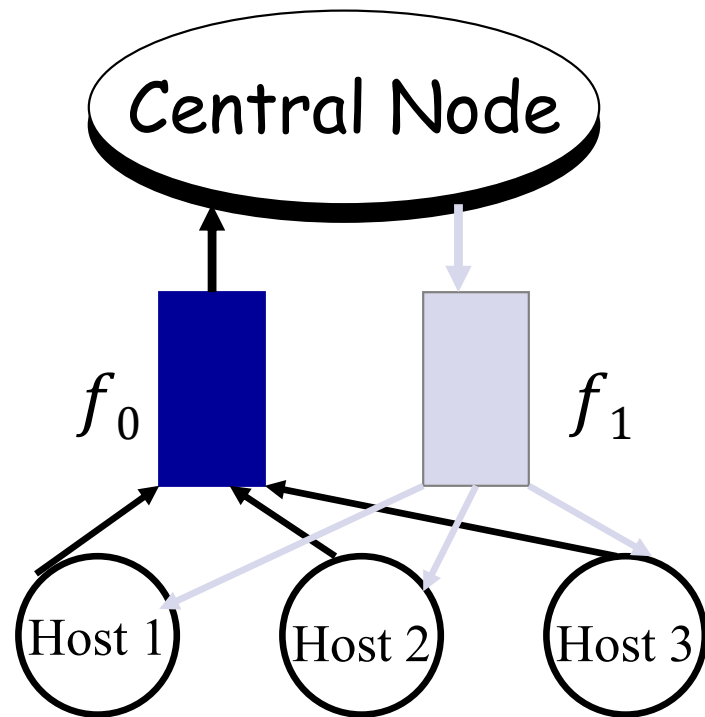
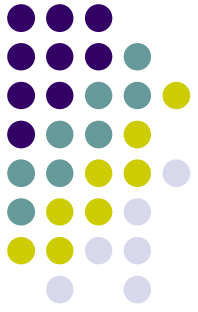
Example: bus arbiter in a computer
What's wrong with that solution?

Alternative 2: Go wild ALOHA



- Whenever you want to send, send
- May have collisions!
 - must verify that message was received
 - use error-detection code
 - receiver sends acknowledgment if all is fine
 - sender waits for ACK (RTT)
- If no ACK:
 - Wait some time and resend
 - Abort if too many failures.

Aloha Protocol (U.Hawaii, '70)

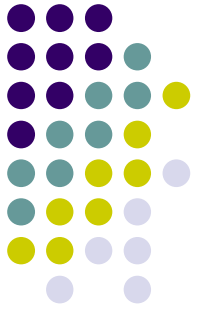


Basic operation:

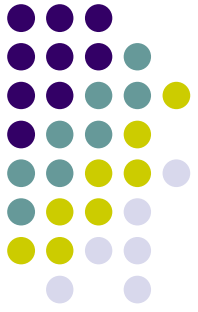
1. All hosts
 - transmit on f_0
 - receive on f_1
2. Central node *repeats* on f_1 whatever it receives on f_0

Collision detected by senders: receive corrupted data
no need for ACK

Aloha Protocol



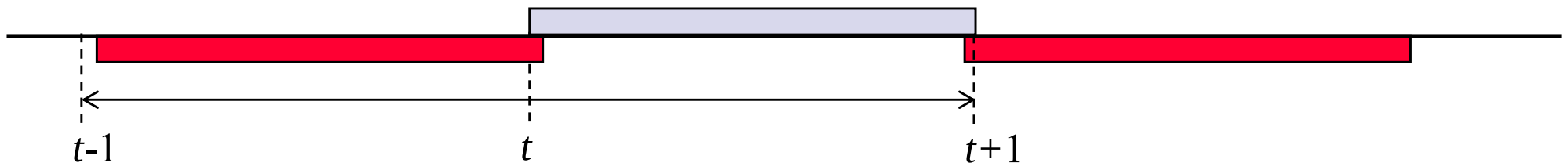
- + Simple
- + Robust against failure of a host.
- + Distributed.
- + Low load $\Rightarrow$ low delay.
- High load $\Rightarrow$ many collisions $\Rightarrow$ low utilization, high delay

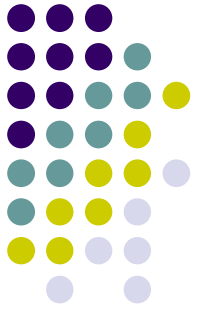


Analysis of ALOHA

Assume transmitting a frame takes one unit of time

- Focus on a collision.
- Which transmissions may collide with a transmission that starts at time t ?





Analysis of ALOHA (cont.)

Suppose there are n stations, and the probability that a station starts transmitting in a time unit is p .

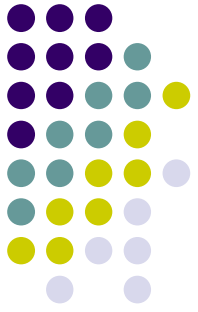
Then: The probability that exactly one node transmits in a time unit is

$$n p ((1 - p)^2)^{n-1}$$

choices of who transmits

Prob. it transmits in the time unit

Prob. all others are silent for 2 units

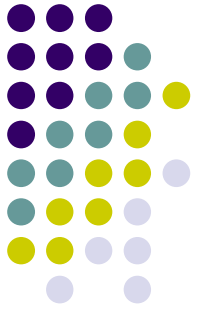


Analysis of ALOHA (cont.)

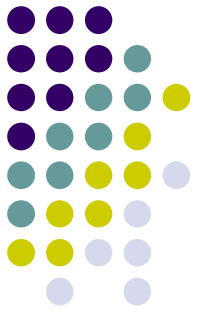
Utilization = fraction of time used for successful transmissions

- Here, the same as $\text{Pr}[\text{time unit successful}]$
- Want to maximize utilization:
 - Differentiate $np((1 - p)^2)^{n-1}$ wrt p
- Get maximum at $p = 1/(2n - 1)$
 - And then utilization is about $\frac{1}{2e} \approx 0.18$.
 - Ouch!

Simple & effective improvement: slotted ALOHA



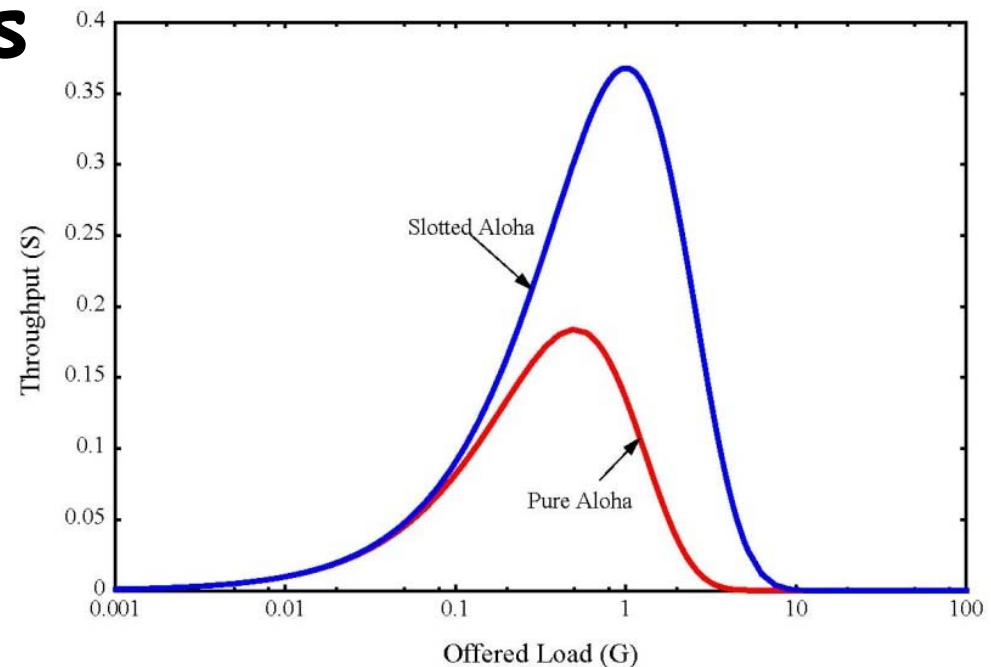
- Why be vulnerable for 2 time units?
- Synchronize, and use **slotted** time:
 - May transmit only at integer times
- Utilization formula now becomes
$$np(1 - p)^{n-1}$$
- Max at $p = 1/n$
 - utilization then is $\frac{1}{e} \approx 0.37$.
- Not that bad...

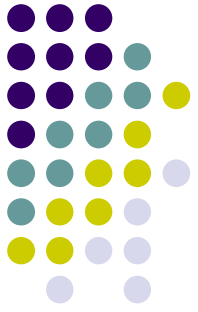


What if link is overloaded?

"Congestion collapse", "network meltdown":

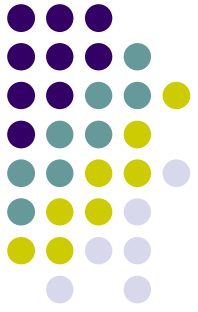
- When offered traffic increases beyond max point ($1/e \approx 37\%$), then total throughput **decreases**
- Retransmissions make things even worse...



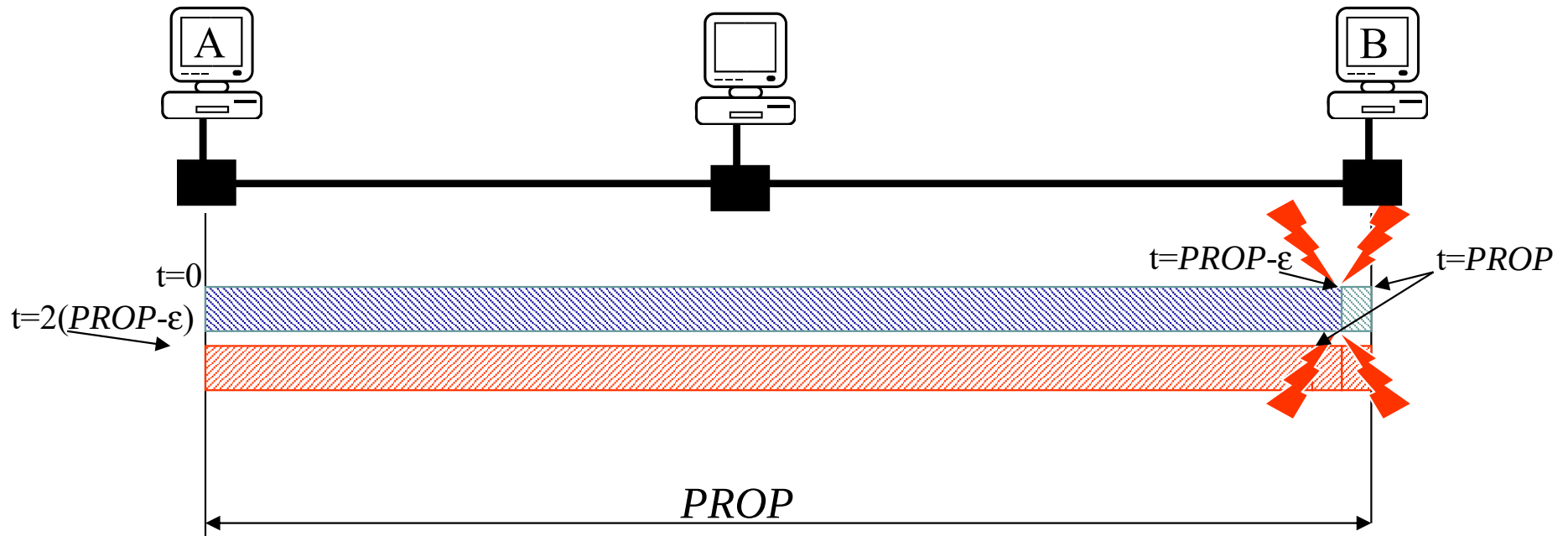


Next step: CSMA/CD

- Hard to synchronize clocks...
- Assume link allows **listening while transmitting**
 - Simple with wires, not with radio
- Protocol: listen always; wait for silence to transmit; abort on collision
- **Carrier Sense Multiple Access / Collision Detection**

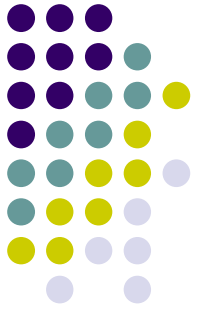


CSMA/CD Size Restriction



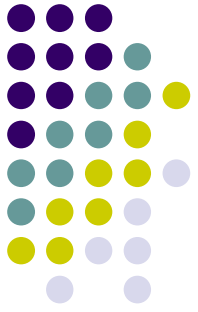
Requirement: detect a collision **before** finishing the transmission.

Must have: minimum packet length $\geq 2 \cdot PROP$



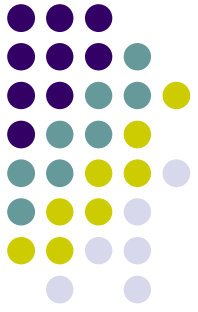
Collisions in CSMA/CD

- Analysis similar to slotted ALOHA, if all transmissions are at least $2 \cdot \text{prop}$ time units.
- Best utilization: when probability of transmission is $1/n$
 - Best utilization is then $\sim 1/e$
- Therefore: when n transmissions collide, let each transmit with probability $1/n$
 - Each waits i time units, i random in $\{0, \dots, n - 1\}$



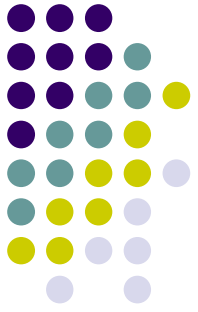
Need $1/n$, but what's n ?

- When transmitting on a CSMA/CD channel, can know if the number of transmitters is
 - More than 1: collision
 - Less than 1: silence
 - Exactly 1: success
- Define abstract problem: Find the value of an unknown on integer n using comparison queries only
 - given value a , answers are only " $>$ ", " $<$ ", " $=$ "

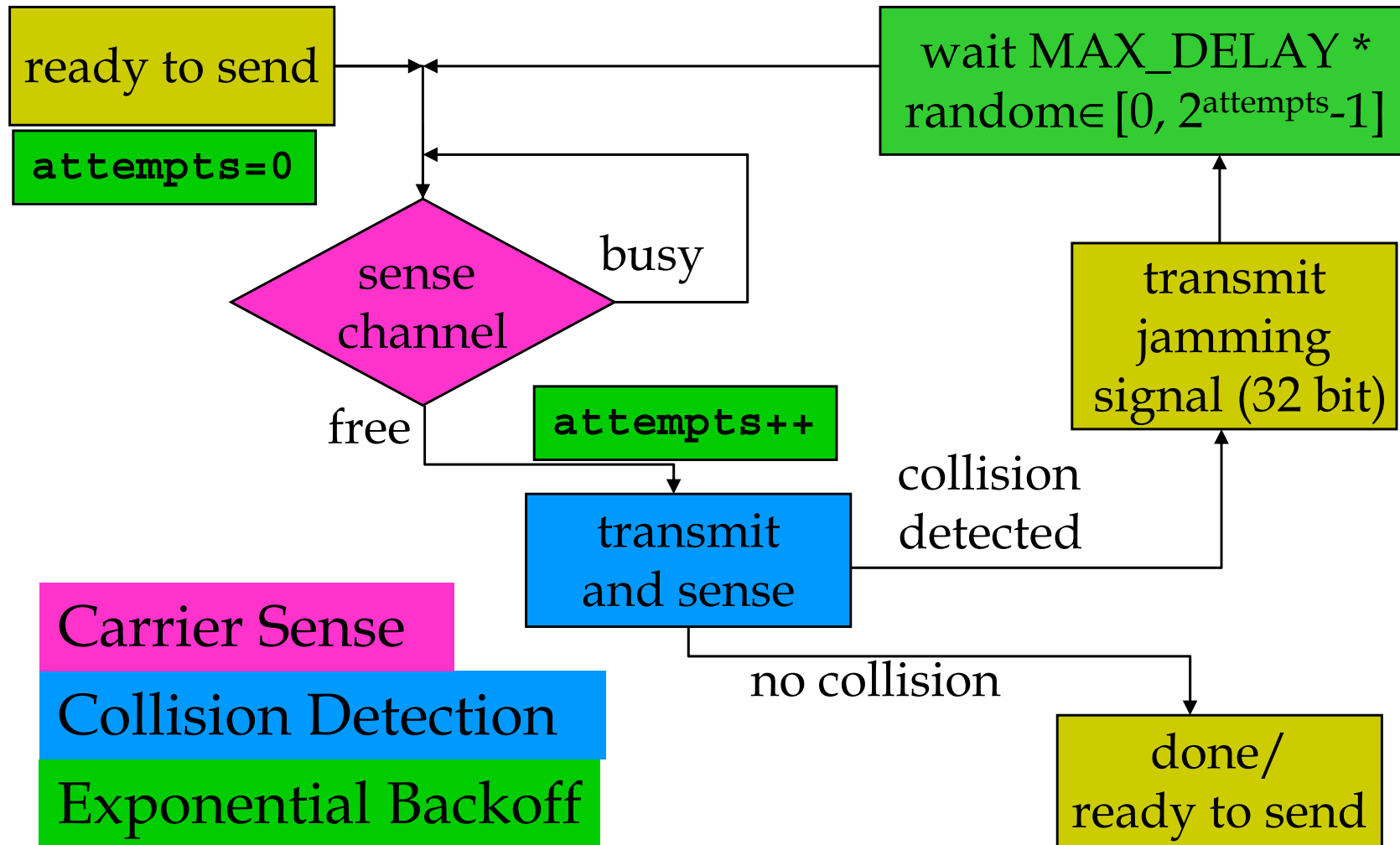


Finding n

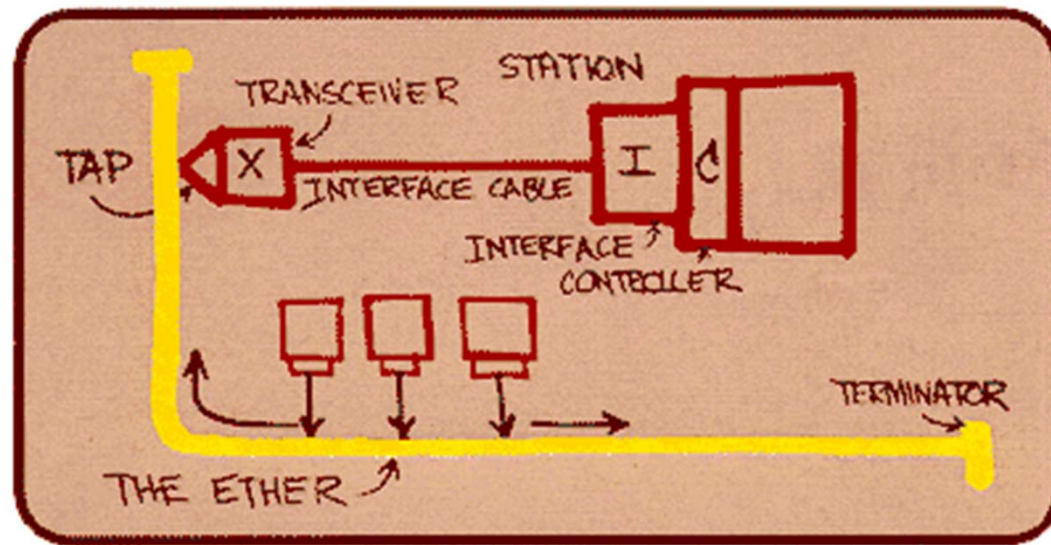
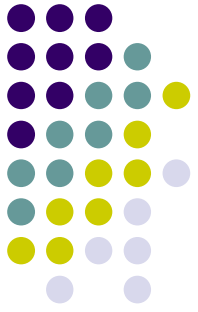
- If we had an upper bound N on n :
 - Use binary search: $\log N$ queries
 - Need to have $N = O(n)$
- How to find such an upper bound N ?
 - Use **exponential backoff**: guess 1,2,4,8...
 - Get a bound $N/2 \leq n < N$
 - At most $\log(2n) = 1 + \log n$ queries to find N
- In CSMA/CD: use guess of N
 - How much expected time until success?



802.3 CSMA/CD Algorithm

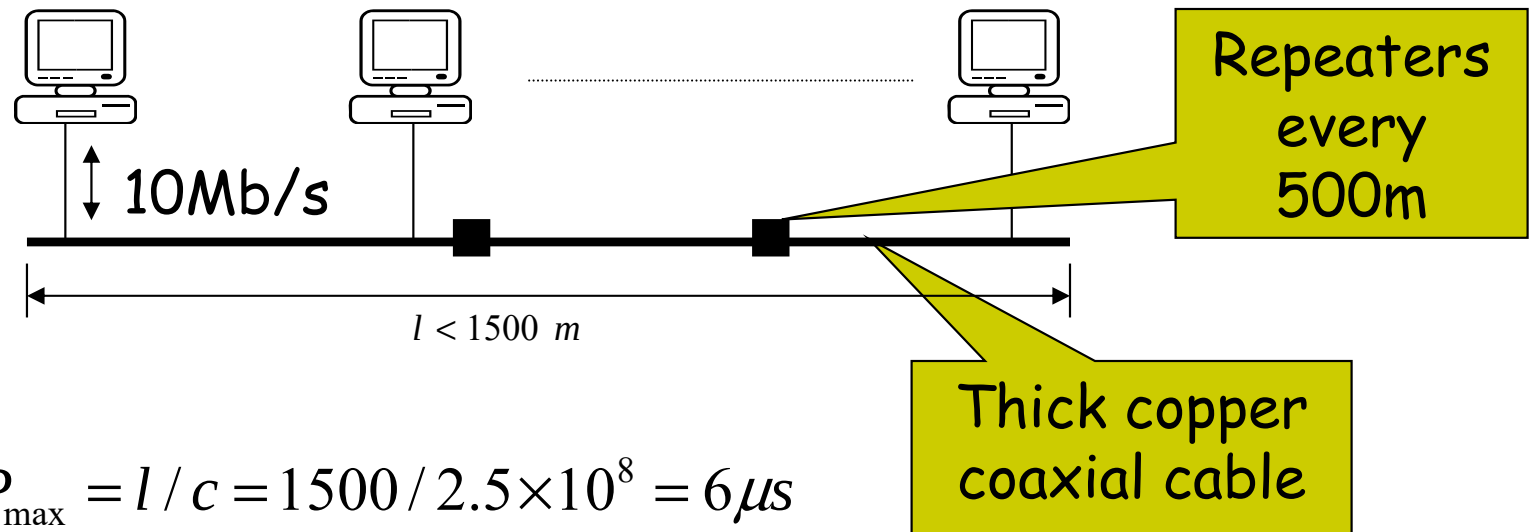
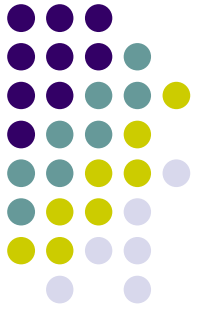


The Original Ethernet



Original picture drawn by Bob Metcalfe,
inventor of Ethernet (1972 - Xerox PARC)

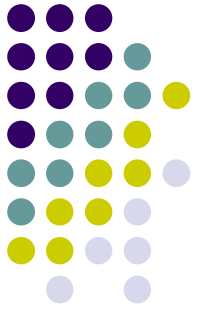
The Original Ethernet



$$PROP_{\max} = l / c = 1500 / 2.5 \times 10^8 = 6 \mu s$$

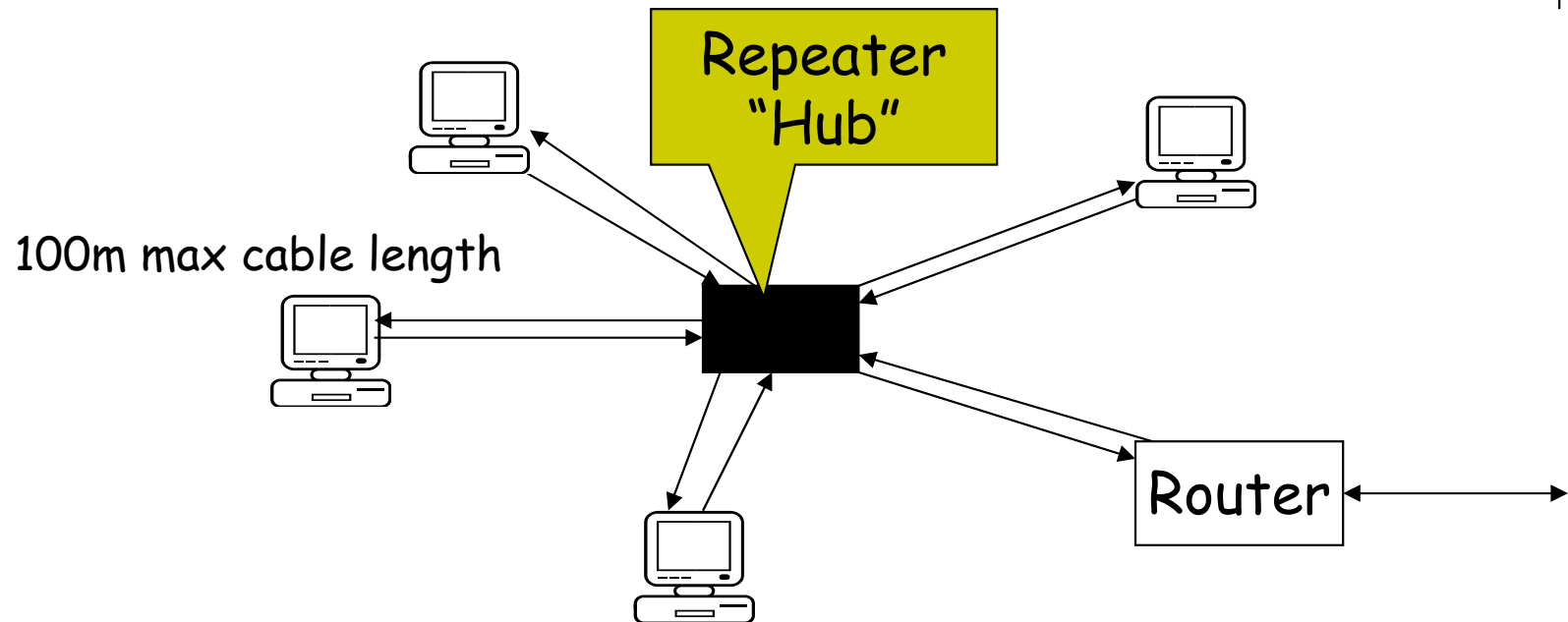
$$\Rightarrow \text{Packet size} \geq (12 \mu s) \times 10 \text{ Mb/s} = 120 \text{ bits}$$

Eventually, minimum packet size is 64 bytes = 512 bits.

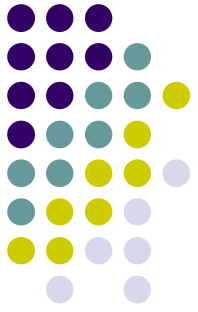


10Base-T

"Twisted pair Ethernet"



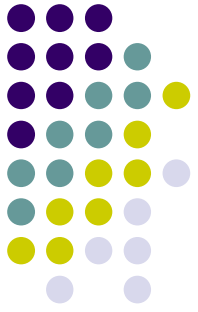
- Runs over twisted pair telephone wire ("cat-3").
- Centralized management ("managed hubs"): better reliability.
- Extremely successful



Ethernet Frame Format

Bytes:	7	1	6	6	2	0-1500	0-46	4
	Preamble	SFD	DA	SA	Type	Data	Pad	CRC

- Preamble: synchronize clocks (7 x 10101010)
- Start of Frame Delimiter: 10101011
- Destination Address: 48-bit globally unique address assigned by manufacturer.
- Type: Indicates protocol of inside layer (IP = 0x0800)
- Pad: Zeroes used to ensure minimum frame length of 64B
- Cyclic Redundancy Check: check sequence to detect bit errors.



Faster Ethernets

Problem: must keep minimal packet size.

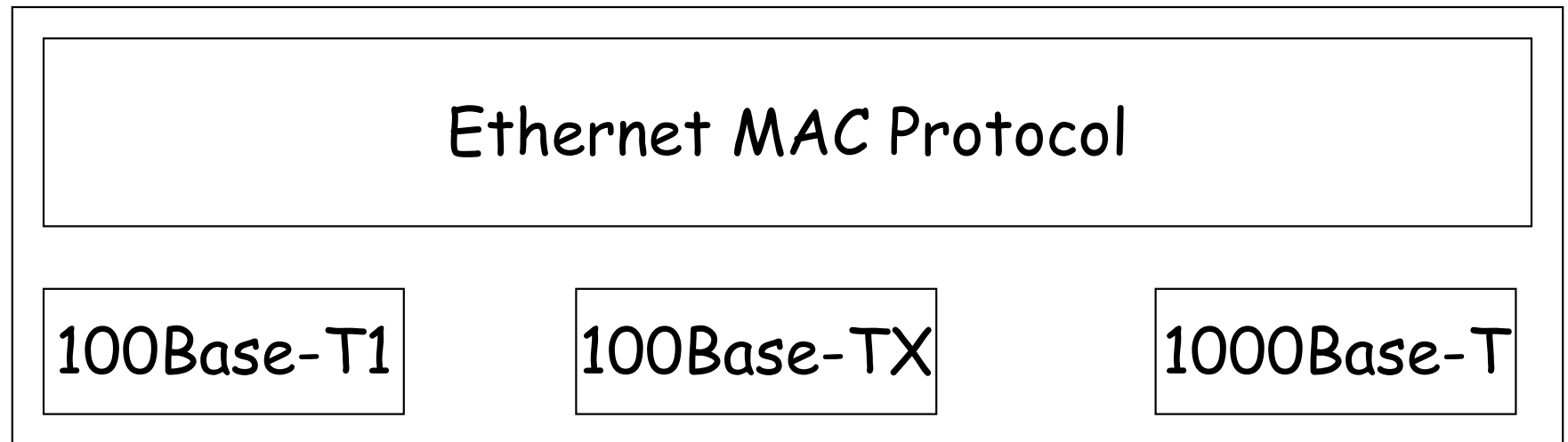
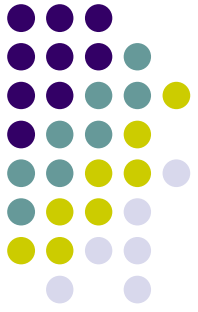
With fixed cable length and propagation speed, must increase minimal size proportionally to bandwidth increase!

- E.g. 100Mb/s, 1500m of cable: prop remains $6\mu\text{s}$, minimal size becomes 1200 bits.

Solutions:

- Cable length limited to 100m.
- Avoid collisions by "Ethernet Switches" (later).

Some Ethernet Standards



Up to 100m of cable per segment.

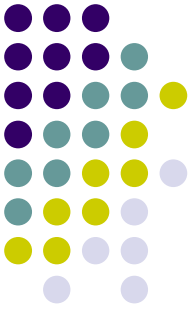
100Base-T1: 1 pair of Cat-5e cable, 15m (cars, IoT).

100Base-TX: 2 pairs of Cat-5 cable, 100m.

1000Base-T: 4 pairs of Cat-5 cable, 100m.

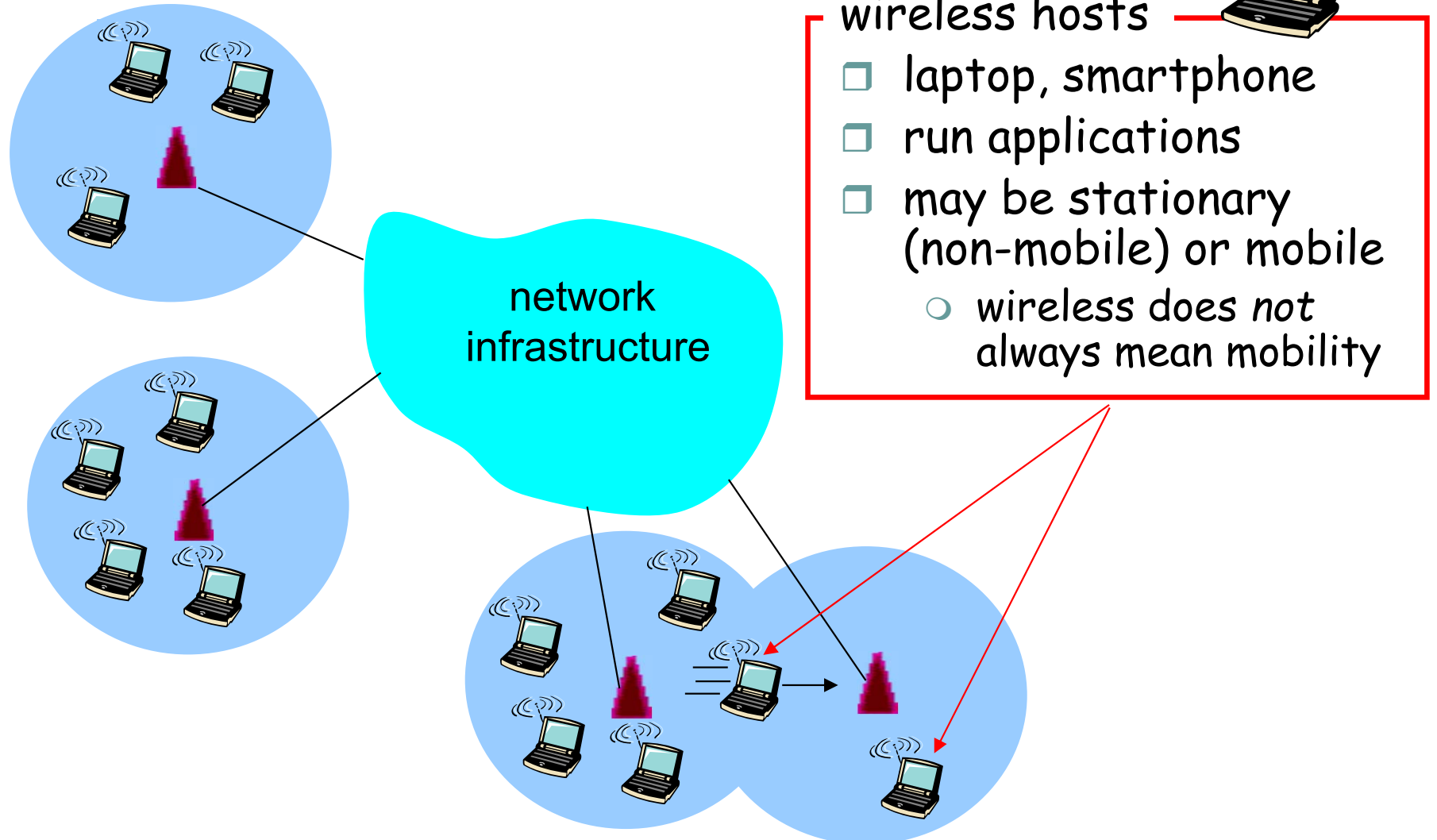
1000Base-?X: 1 pair of fibers, 10's of kilometers

10GBase-*, 40G, 100G, ...

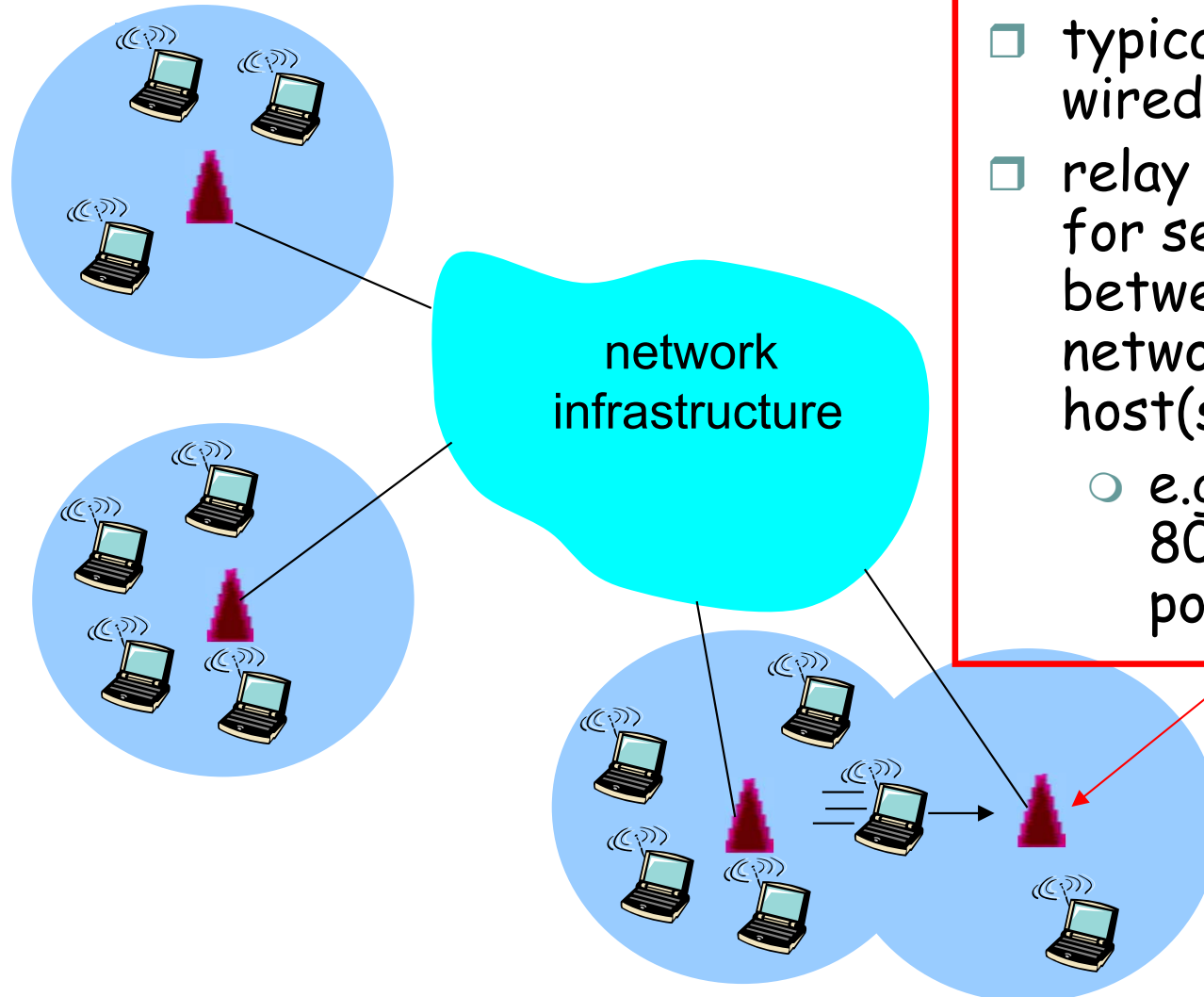


WIRELESS LANS

Elements of a wireless network



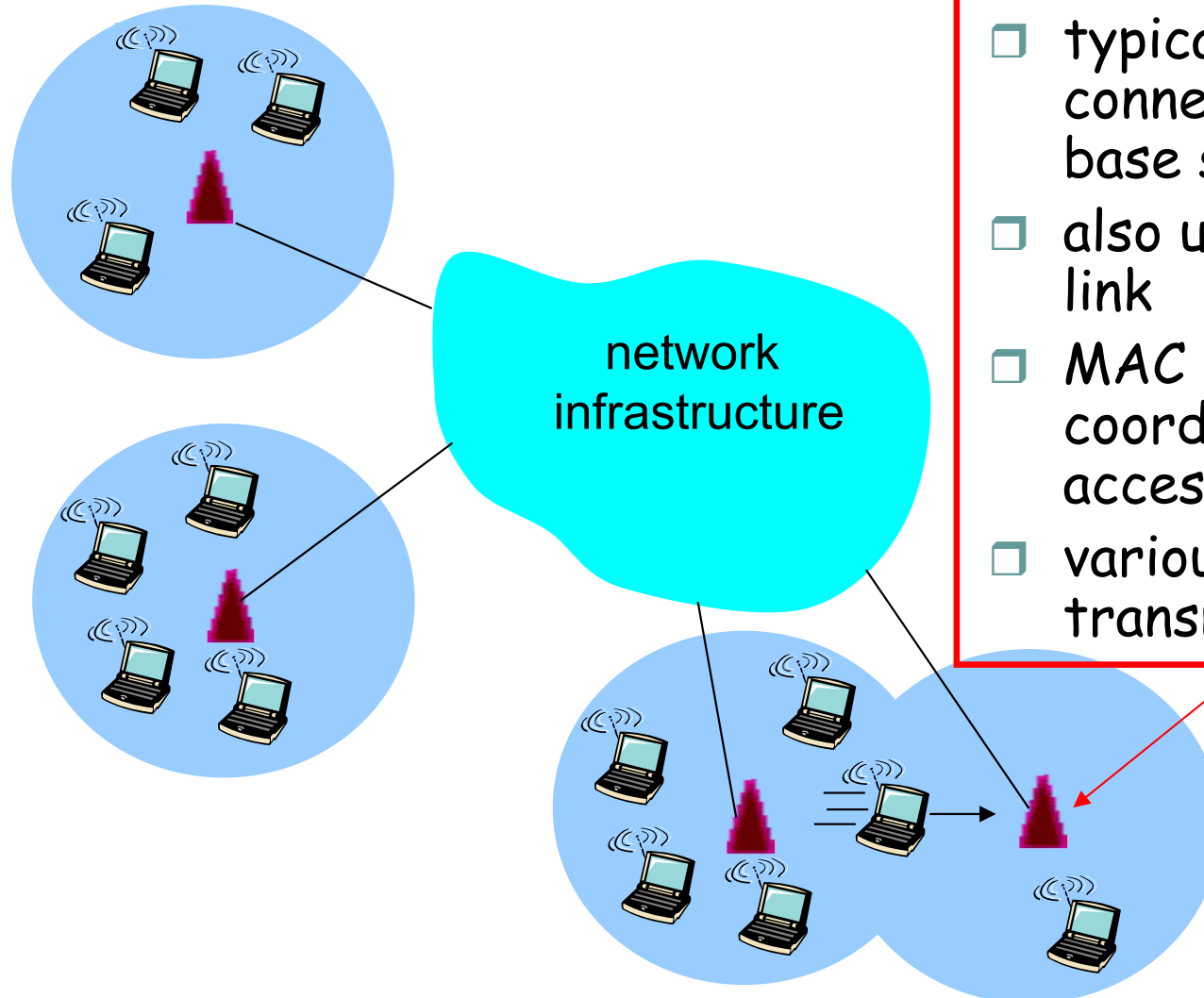
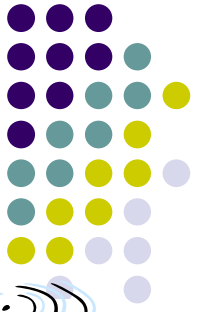
Elements of a wireless network



base station

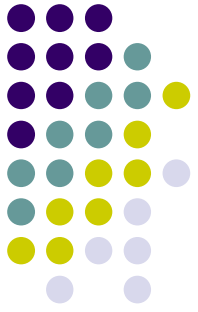
- ❑ typically connected to wired network
- ❑ relay - responsible for sending packets between wired network and wireless host(s) in its "area"
 - e.g., cell towers, 802.11 access points

Elements of a wireless network

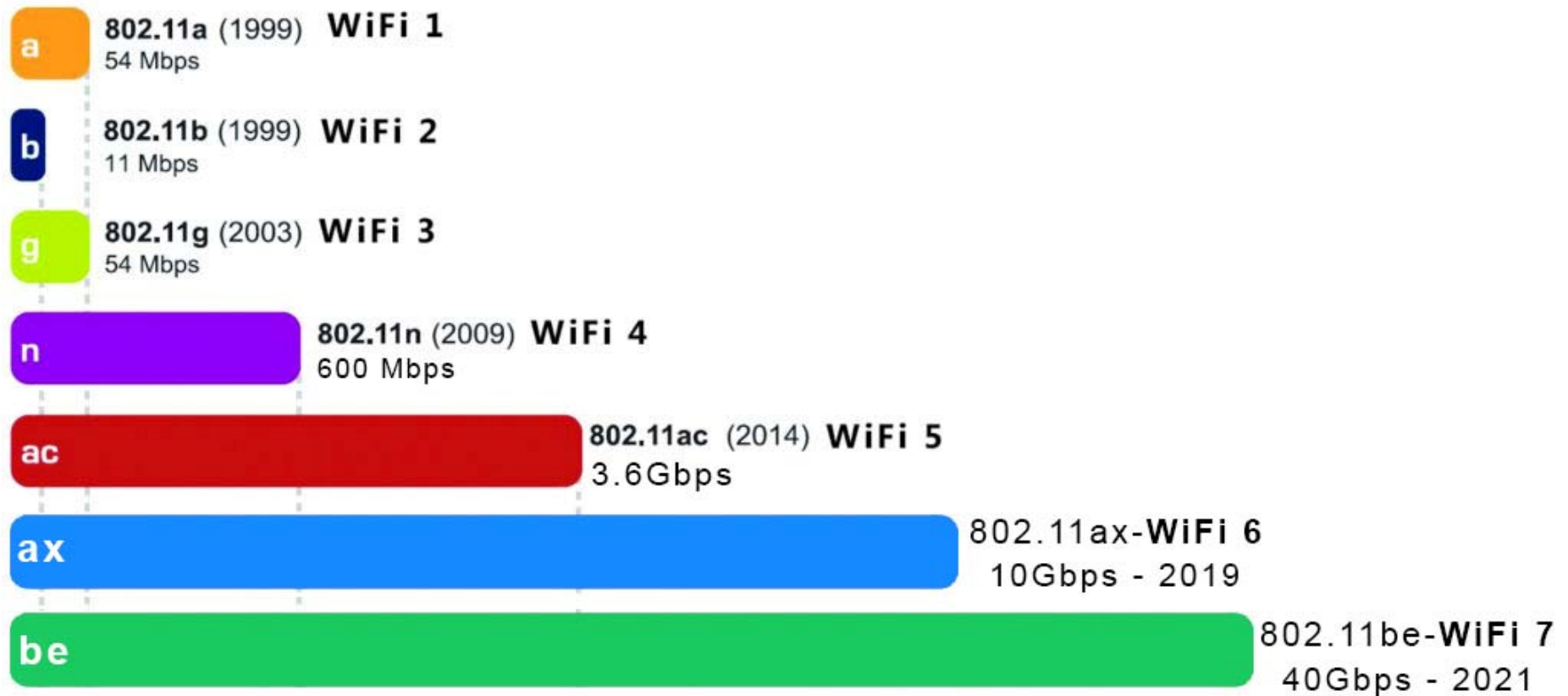


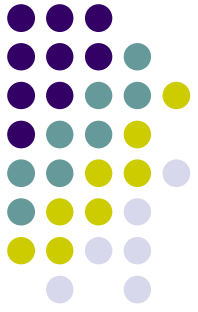
wireless link

- ❑ typically used to connect mobile(s) to base station
- ❑ also used as backbone link
- ❑ MAC protocol coordinates link access
- ❑ various data rates, transmission distance

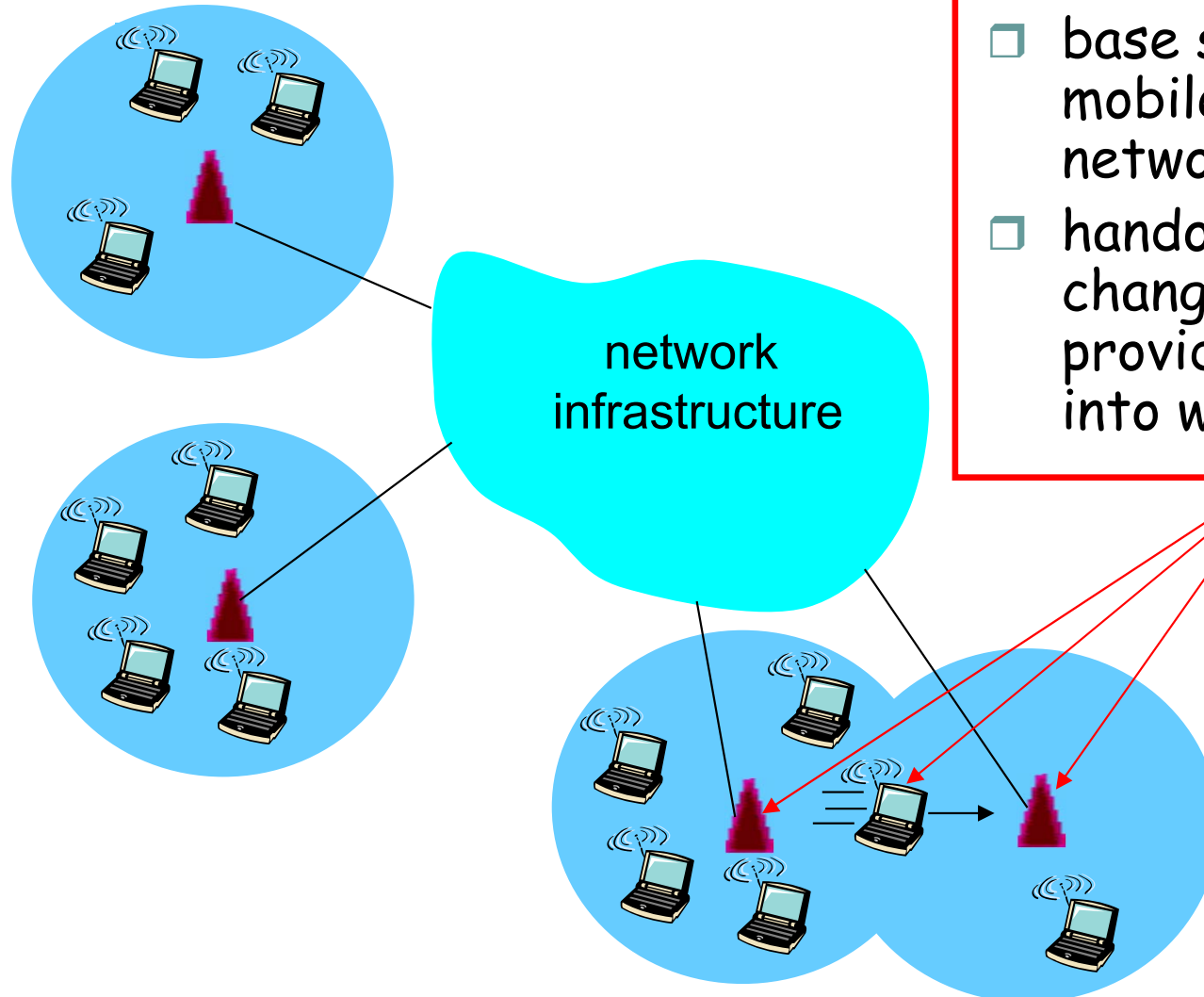


Some wireless link standards



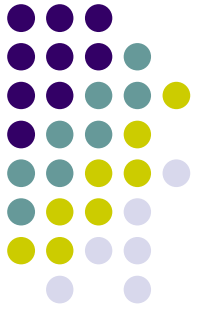


Elements of a wireless network

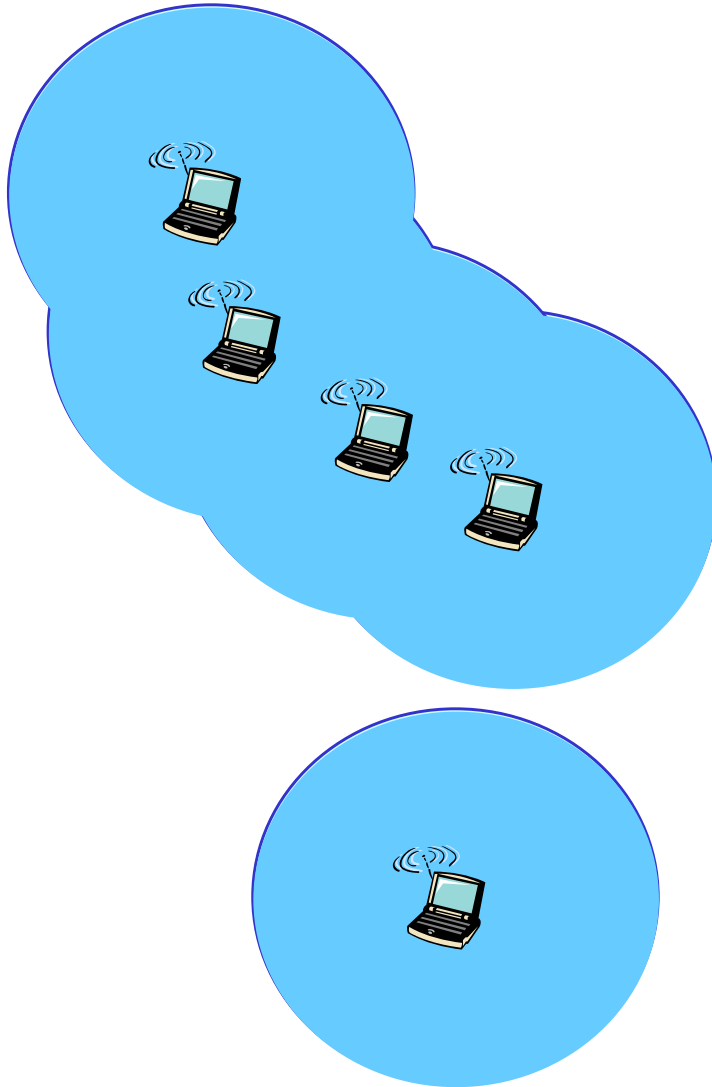


infrastructure mode

- ☐ base station connects mobiles into wired network
- ☐ handoff: mobile changes base station providing connection into wired network



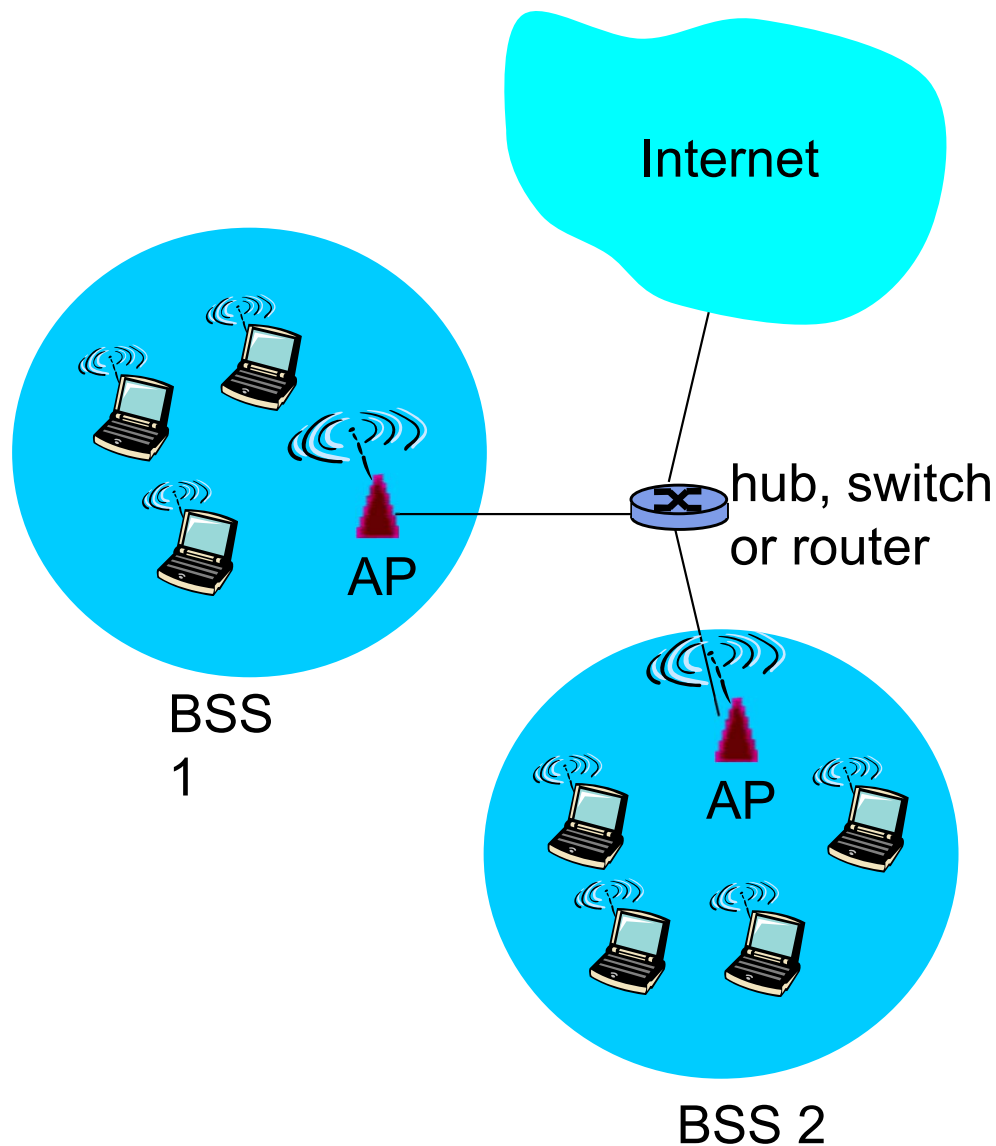
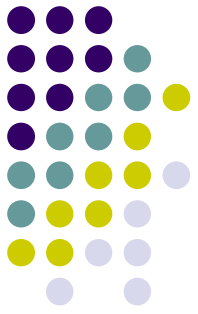
Elements of a wireless network



ad hoc mode

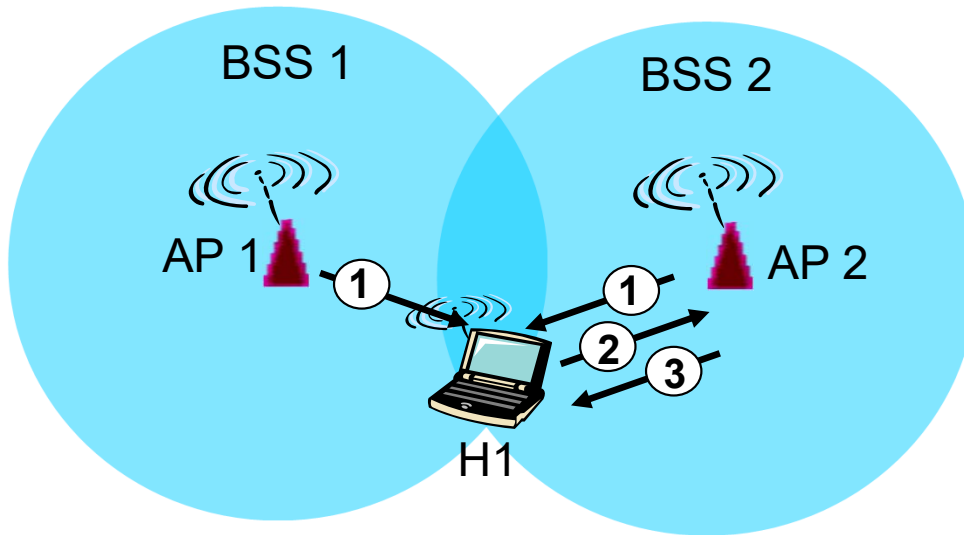
- ☐ no base stations
- ☐ nodes can only transmit to other nodes within link coverage
- ☐ nodes organize themselves into a network: route among themselves

802.11 LAN architecture



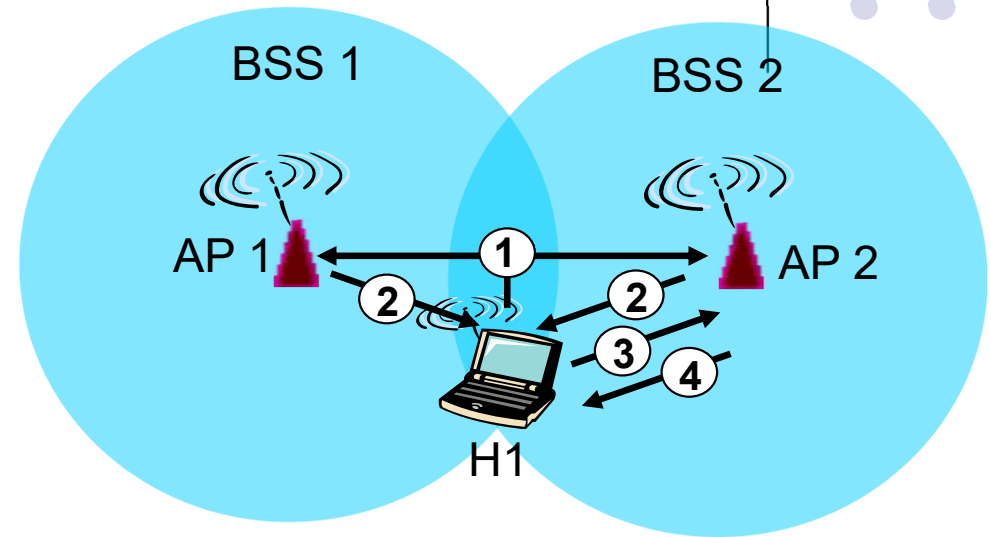
- ❑ wireless host communicates with base station
 - base station = access point (AP)
- ❑ Basic Service Set (BSS) (aka "cell") in infrastructure mode contains:
 - wireless hosts
 - access point (AP): base station
 - ad hoc mode: hosts only

802.11: passive/active scanning



Passive Scanning:

- (1) beacon frames sent from APs
- (2) association Request frame sent:
H1 to selected AP
- (3) association Response frame sent:
H1 to selected AP

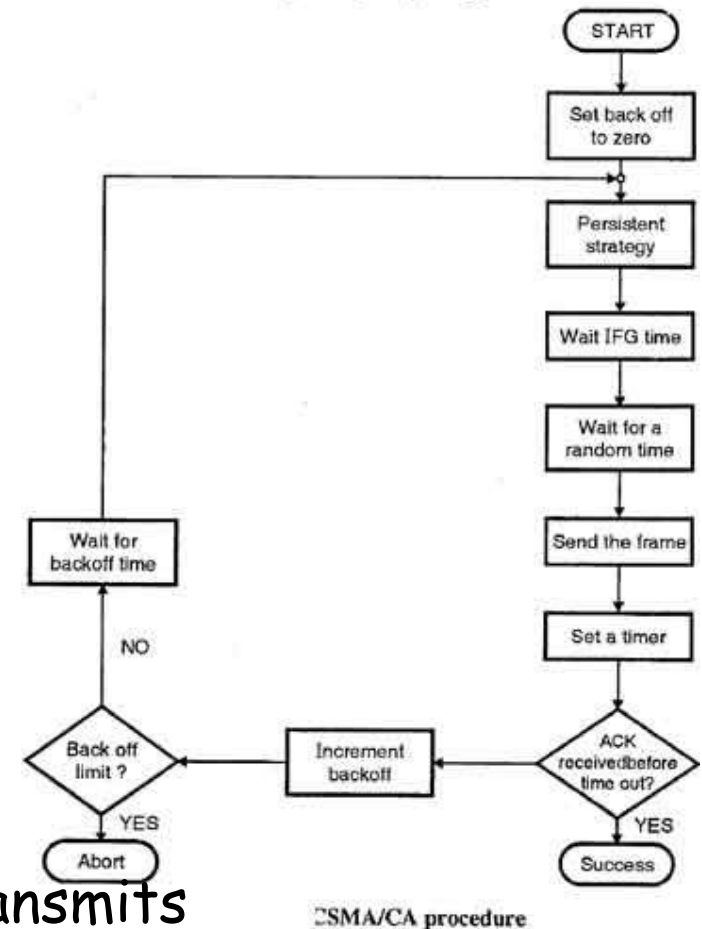


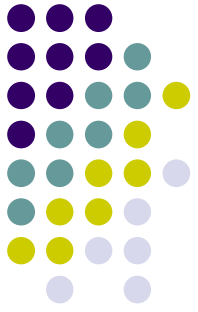
Active Scanning:

- (1) Probe Request frame broadcast
from H1
- (2) Probes response frame sent from
APs
- (3) Association Request frame sent:
H1 to selected AP
- (4) Association Response frame
sent: H1 to selected AP

Wireless LANs MAC: CSMA/CA

- Radio xmit/recv power ratio is huge
→ Can't **detect** collision
- Many bit errors
- Consequences:
 - Use explicit acks
 - Try to avoid collisions
- CSMA/CA Algorithm:
 - Wait until silence
 - Set timer to $\text{random} \in [1, CW]$
 - Pause timer while another station transmits
 - On timeout, send packet and wait for ack
 - If no ack: $CW \leftarrow 2CW$, try again



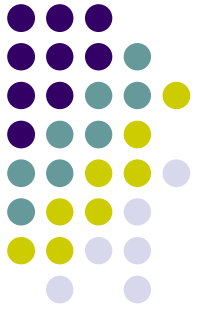


The problems of asymmetry

CSMA/CA doesn't work when not all stations can receive transmissions from all stations

- Hidden terminal
 - some stations in an area cannot hear transmissions from others, though base can hear both
 - collision because carrier not detected
- Exposed terminal
 - some (but not all) stations can hear transmissions from stations not in the local area
 - idle station because carrier incorrectly detected

Hidden terminal



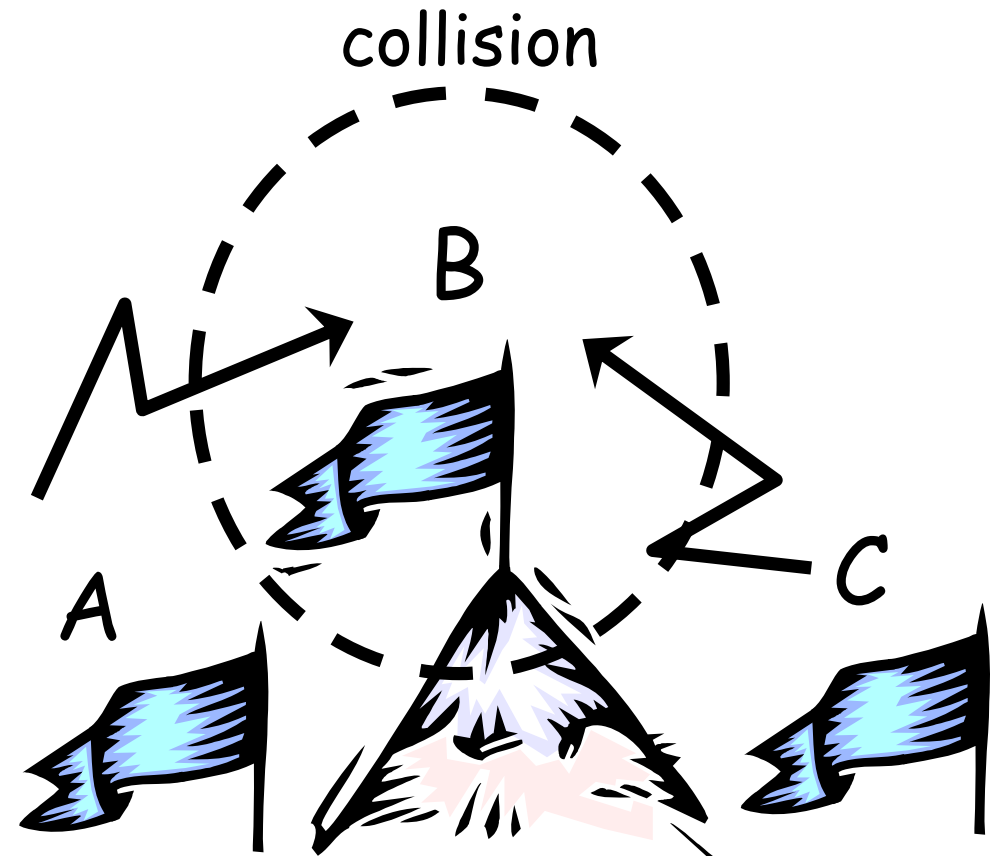
- B can hear both A and C
- A and C can't hear each other
- When A transmits to B, C may interfere

Notation: $A \dashrightarrow B$ means that B can hear A.

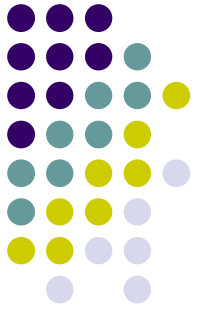
$A \dashleftrightarrow B \dashleftrightarrow C$ doesn't imply $A \dashleftrightarrow C$

Hidden terminal:

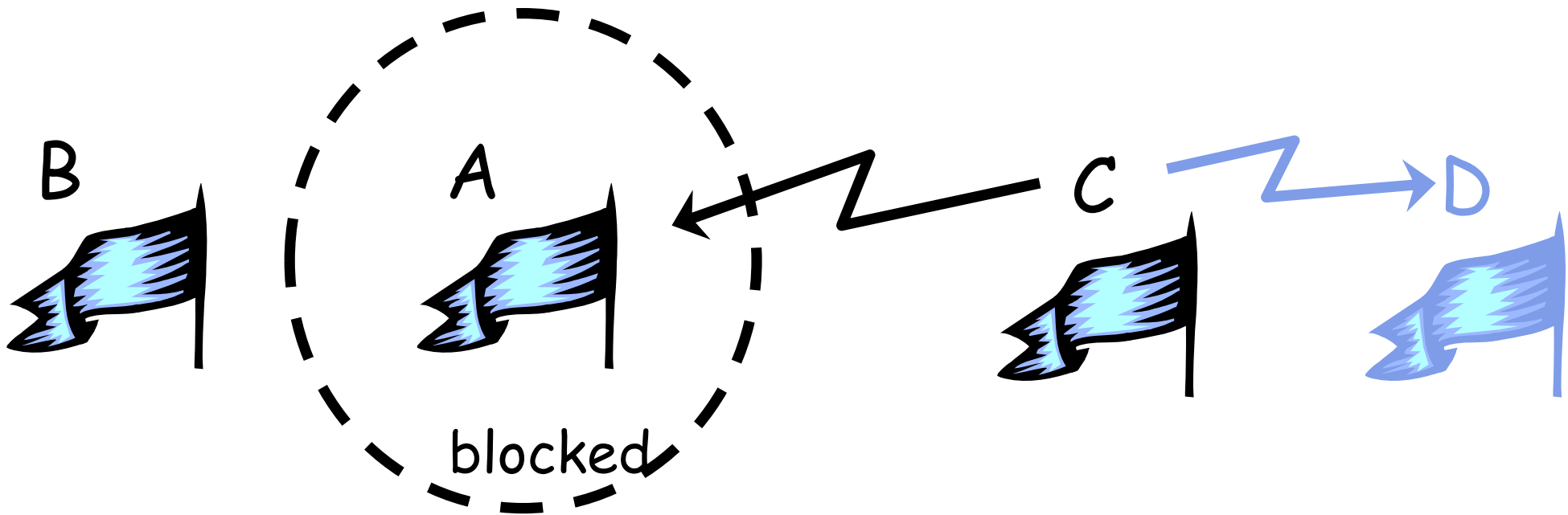
$A \dashrightarrow B, C \dashrightarrow B, A \not\leftrightarrow C$



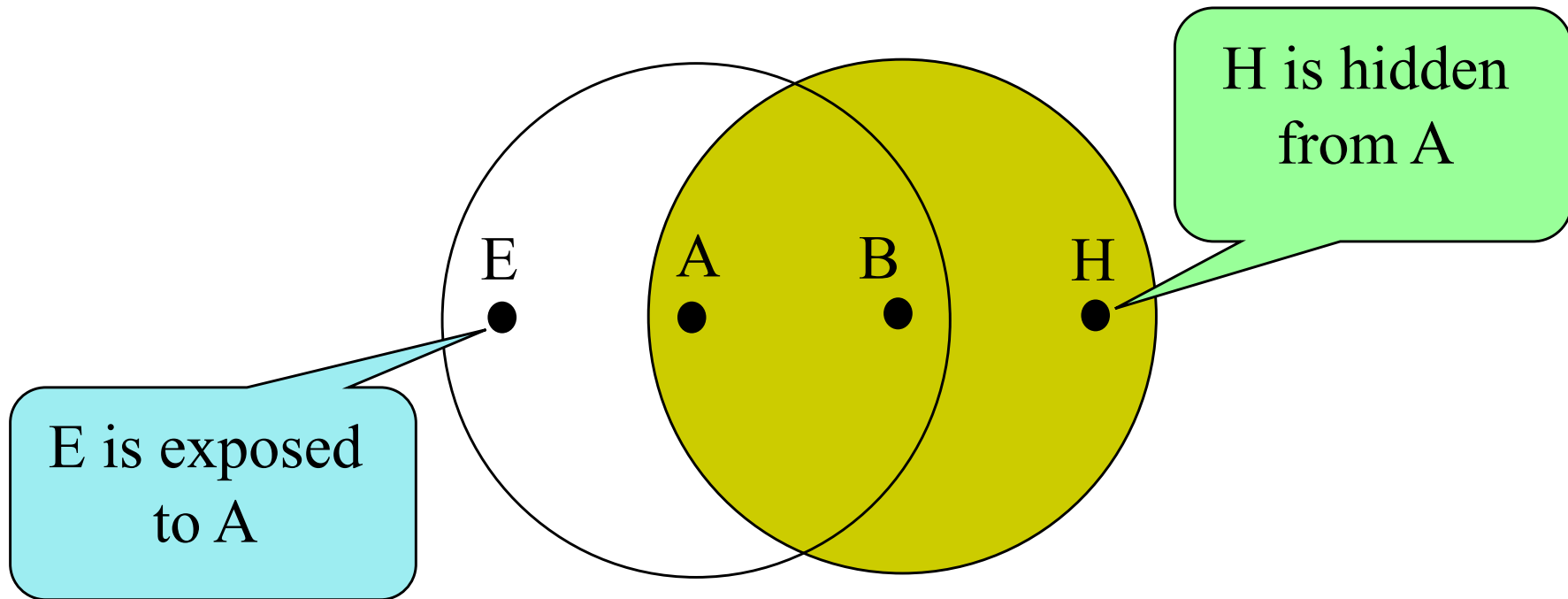
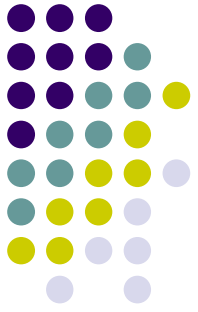
Exposed terminal



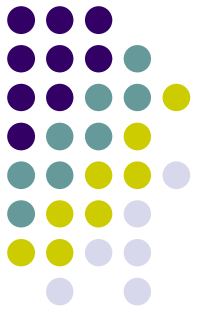
- Assumption: $B \not\rightarrow C$, but $A \rightarrow C$; $D \not\rightarrow A$
- Result: When C transmits to D , A does not transmit to B even though it could!
- Exposed terminal: $A \rightarrow B$, $C \not\rightarrow B$ does not mean $C \not\rightarrow A$.



The problems summarized

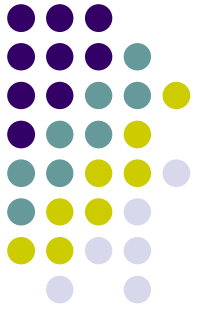


Dealing with hidden and exposed terminals



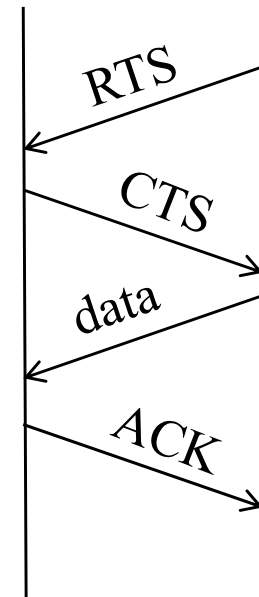
- Busy Tone Multiple Access (BTMA)
 - uses a separate "busy-tone" channel
 - when X is receiving a message, X transmits on BT channel
 - everyone who might want to talk to X knows that X is busy even if they cannot hear transmission to X
 - this avoids both problems (why?)
- BTMA requires us to split frequency band
 - more complex receivers (need two tuners)
 - If bands may have different propagation characteristics
-> scheme fails!

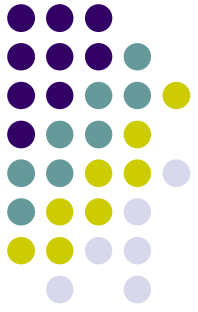
Multiple Access Collision Avoidance (MACA)



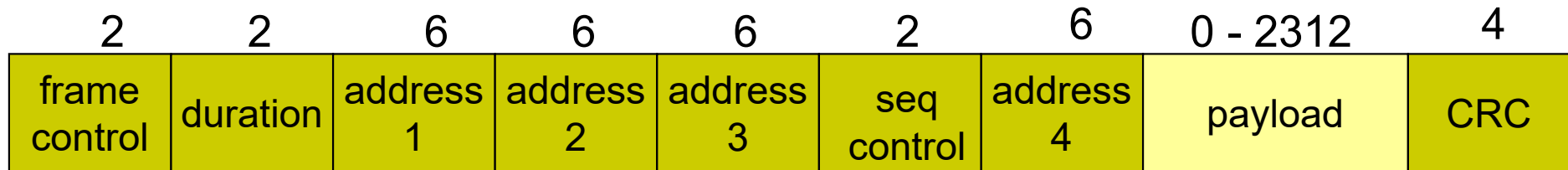
Optional in 802.11

- Sender transmits `RequestToSend` (RTS) frame
- Receiver replies with `clearToSend` (CTS) frame
- Neighbors...
 - see CTS: keep quiet
 - see RTS but not CTS: ok to transmit
- Receiver sends `ACK` when has frame
 - neighbors silent until see `ACK`
- Collisions
 - no collisions detection
 - known when don't receive CTS
 - exponential backoff





802.11 frame: addressing



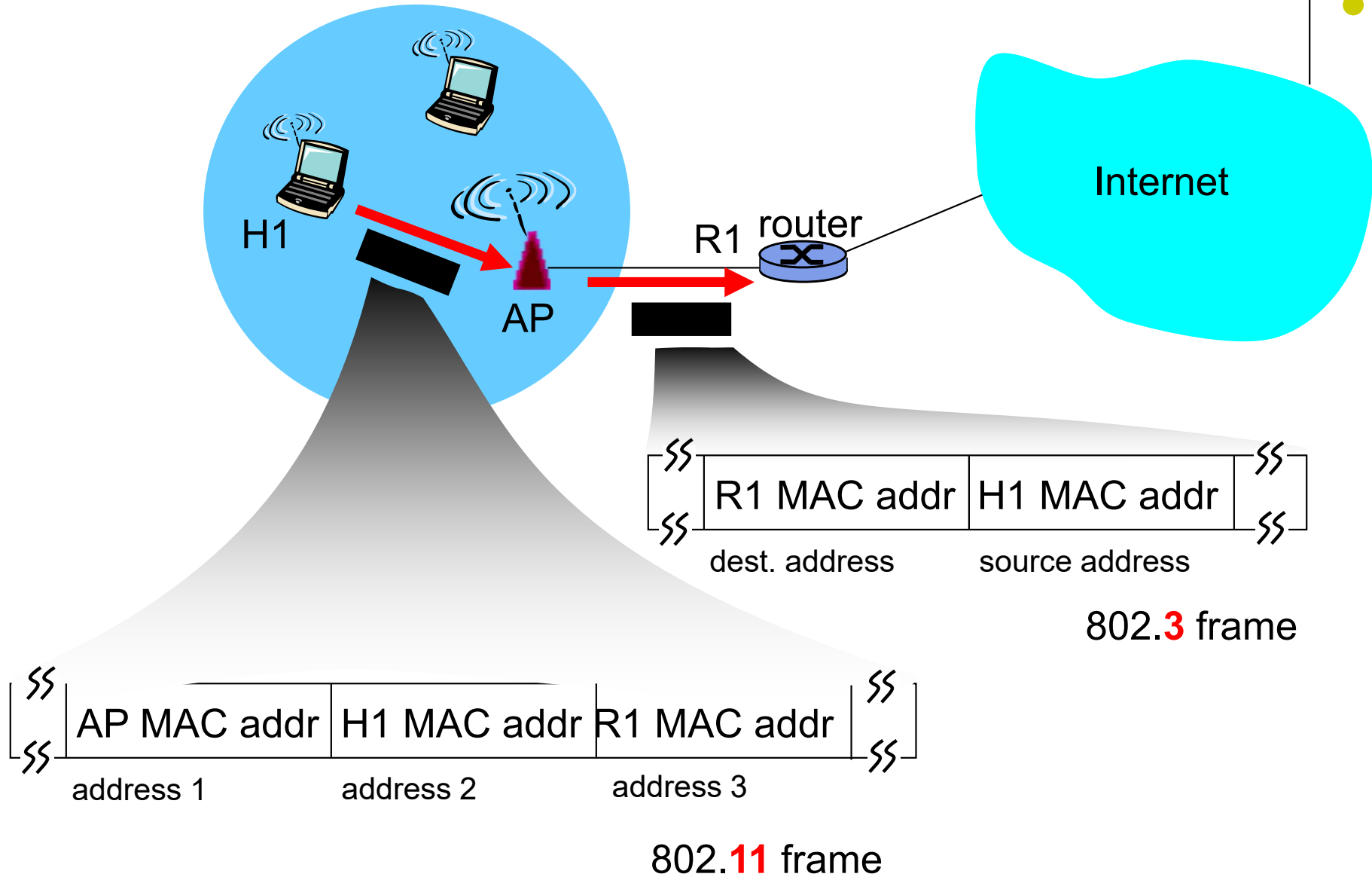
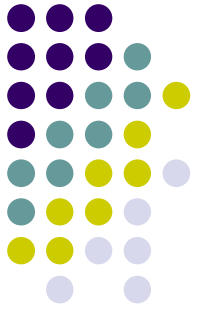
Address 1: MAC address of wireless host or AP to receive this frame

Address 2: MAC address of wireless host or AP transmitting this frame

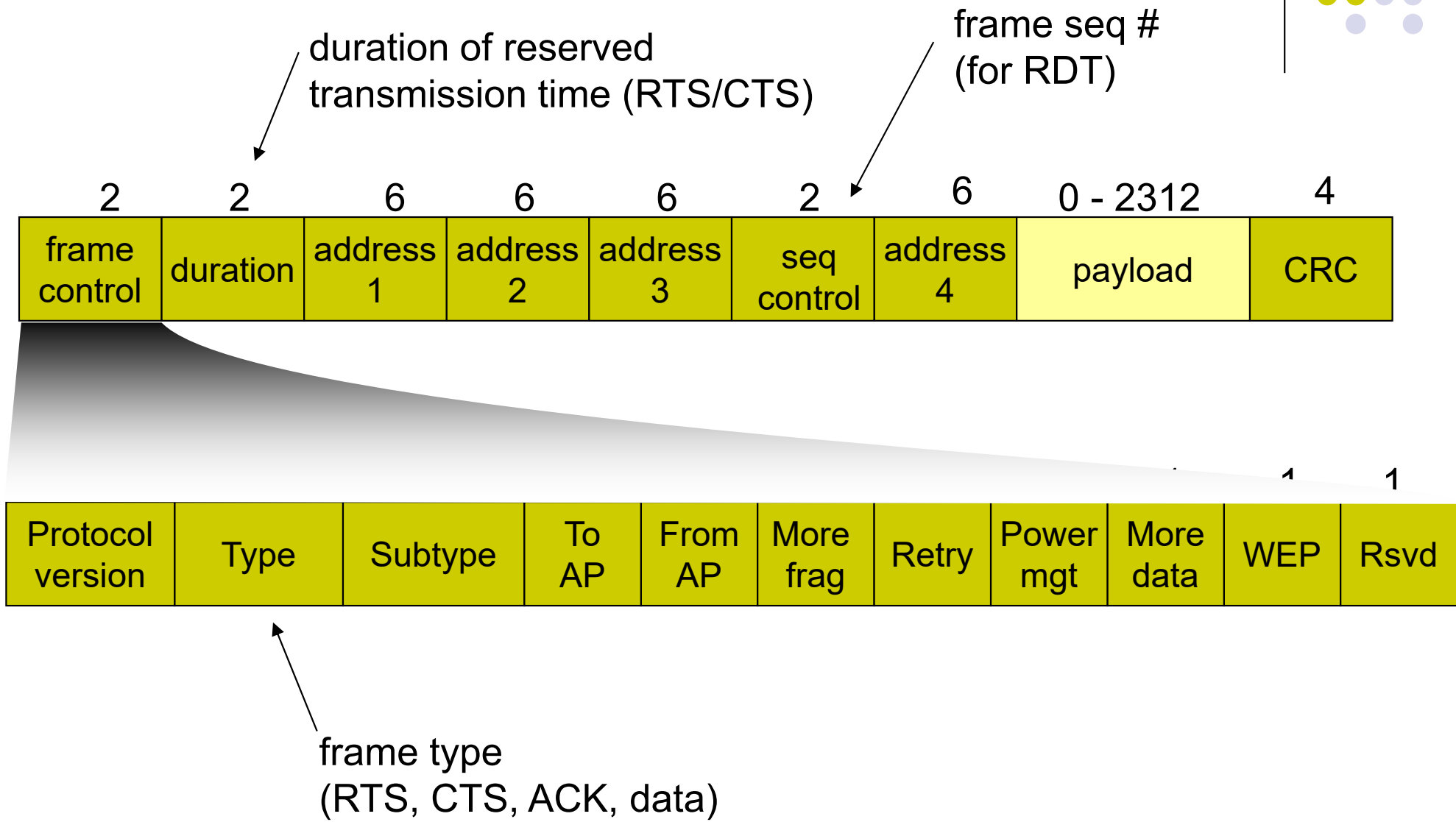
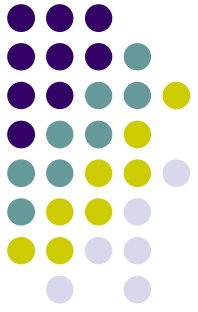
Address 3: MAC address of router interface to which AP is attached

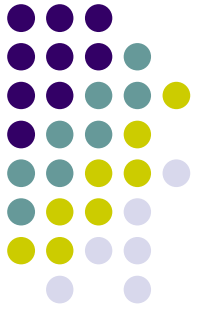
Address 4: used only in ad hoc mode

802.11 frame: addressing



802.11 frame: more





New Concepts

- TDMA, FDMA
- Little's Theorem
- Arrival process
- M/M/1 model
- Markov Chain
- Steady state
- Balance equations
- Aloha
- Slotted Aloha
- Collision detection
- Exponential backoff
- Ethernet IEEE 802.3
- Collision avoidance
- Hidden terminal
- Wifi IEEE 802.11
- MACA